Mohamed Mahrous
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EDUCATION

- **PhD**, Electrical and Computer Engineering — **Jan 2024 – Dec 2027**
  University of Toronto, Canada
- **Master of Science with Honors, Nanotechnology** — **Feb 2020 – July 2023**
  The American University in Cairo (AUC), Egypt
- **Bachelor of Science, Chemistry and Physics** — **Sep 2012 - May 2016**
  University of Sohag, Egypt

WORK EXPERIENCE

- **AI Application Engineer (Internship), Pulsenics, Canada, June.2025 – Oct 2025**
  - Developed machine learning algorithms for real-time electrochemical diagnostics using in-operando EIS data.
  - Designed a predictive framework to analyze PEM fuel cell stack performance across 27 operating conditions.
  - Applied regression and time-series models to correlate impedance features with system degradation.
- **Graduate Teaching & Research Assistant, University of Toronto, Canada, Jan.2024 – Dec.2027**
  - Teaching ECE221H1, MAT291H1, and ECE191.
  - Using atomistic modeling and Machine Learning to study interfacial electrochemical reactions in battery systems.
- **Visiting Research Scholar, Johns Hopkins University, Baltimore - USA, Jul.2023 – Dec.2023**
  - Worked on Machine Learning design of High Entropy Alloys.
- **Assistant Lecturer, Universidade Nova de Lisboa (part time), Cairo - Egypt. Sep.2022 – Dec.2022**
  - Delivered lectures and facilitated student learning for the Foundational Chemistry course.
- **Research assistant, EML Laboratory-Computational subgroup, The American University, Cairo - Egypt**. **Nov.2018 – Jul.2023**
  - Performed computational modeling to develop novel materials with tailored properties.
  - Conducted ab initio simulations of molecular–surface interactions for catalysis and sensor applications.
- **Teaching Assistant, University of Prince Edward Island, Cairo - Egypt, Oct.2018 – Oct.2020**
  - Taught General Chemistry and Sustainable Design courses.
- **Quality control specialist, Quality Department, PFNonwovens, Cairo - Egypt, Dec.2016 – Oct.2018**
  - Monitored the quality of the products, testing parameters, working procedures, and performing audits on other departments.

Training and Workshops

- **Participant**, American Control Conference (ACC 2024) - American Automatic Control Council, Toronto, Canada **(2024)**
- **Poster Presenter,** Materials Research Society (MRS) Fall Meeting, Boston, USA **(2023)**
- **Poster Presenter,** Green Chemistry and Climate Change Summer School, Ca' Foscari University of Venice, Italy **(2021)**
- **WHMIS 2015 Certification,** University of Prince Edward Island, Canada **(2018)**
- **ISO Systems & 6S Training**, PFNonwovens, Egypt **(2016)**
- **Undergraduate Research Assistant,** Egyptian Center for Theoretical Physics, Egypt **(2015)**

Awards

- University Of Toronto Fellowship, Department Of Electrical And Computer Engineering, **2024 & 2025 & 2026**
- Bert Wasmund Graduate Scholarship for Sustainable Energy Research, **2024**
- Edward S. Rogers Sr. Graduate Scholarship, **2024**
- AUC Honors (Academic Recognition), **2023**
- The American University in Cairo (Laboratory Instructions Fellowship), **2022 & 2021 & 2020**
- The American University in Cairo (University Fellowship), **2023 & 2021**
- The American University in Cairo (Research Grant), **2021**
- The American University in Cairo (Study abroad Grant), **2023**

SKILLS

Languages: English (Fluent), Arabic (Native), French (Intermediate).

Programming & Automation: Python, C, numerical simulations, and machine learning architecture. experience in Autonomous Research Workflows and controlling Self-Driving Labs (SDLs) for rapid material optimization.

Data Analytics & Visualization: Data wrangling, exploratory data analysis (EDA), feature engineering, and scientific visualization using Python (NumPy, Pandas, Matplotlib), and Jupyter Notebooks.

Machine Learning & Informatics: Supervised and unsupervised learning (scikit-learn), regression, classification, clustering, and decision trees. Advanced expertise in Gaussian Processes, Bayesian Active Learning, Multi-Objective Optimization, and Dimensionality Reduction.

Model Validation: Critically evaluating models for overfitting, range of applicability, uncertainty quantification, and physical plausibility in structure property relations.

Computational Modeling: Density Functional Theory (DFT), Molecular Dynamics (MD), and Molecular Mechanics using Material Studio, GULP, Quantum ESPRESSO, LAMMPS, and VASP.

Scientific Software & Tools: OVITO, OriginPro, VESTA, Avogadro, XCrySDen, Microsoft Office Suite.

Quantitative Modeling: Partial Differential Equations (PDEs), state-space models, transport equations, Statistical Design of Experiments (DoE), and Hyperparameter Tuning.

High-Performance Computing: Parallel computing with MPI; experience using HPC clusters and high-throughput simulations

Project Management & Collaboration: Interdisciplinary research coordination, Oral Technical Reporting, and engagement with emerging AI topics including Ethics of AI and Large Language Models (LLMs).

Publications

1. Abbas, W. A., Shaheen, B. S., Ghanem, L. G., Badawy, I. M., **Abodouh, M. M.,** Abdou, S. M., Zada, S., & Allam, N. K. (2021). Cost-Effective Face Mask Filter Based on Hybrid Composite Nanofibrous Layers with High Filtration Efficiency. In Langmuir (Vol. 37, Issue 24, pp. 7492–7502). American Chemical Society (ACS). https://doi.org/10.1021/acs.langmuir.1c00926
2. Sanad, M. F., Puente Santiago, A. R., Tolba, S. A., Ahsan, M. A., Fernandez-Delgado, O., Shawky Adly, M., Hashem, E. M., **Mahrous Abodouh, M.,** El-Shall, M. S., Sreenivasan, S. T., Allam, N. K., & Echegoyen, L. (2021). Co–Cu Bimetallic Metal Organic Framework Catalyst Outperforms the Pt/C Benchmark for Oxygen Reduction. The American Chemical Society (Vol. 143, Issue 10, pp. 4064–4073). American Chemical Society (ACS). https://doi.org/10.1021/jacs.1c01096
3. Elfarargy, R. G., Saleh, M. A., **Abodouh, M. M**., Hamza, M. A., & Allam, N. K. (2022). Graphitic Carbon Nitride Nanoheterostructures as Novel Platforms for the Electrochemical Sensing of the Chemotherapeutic and Immunomodulator Agent MTX. Biosensors (Vol. 13, Issue 1, p. 51). MDPI AG. https://doi.org/10.3390/bios13010051
4. Leil, R., **Abodouh, M. M.,** Javed, N., Sreekumar, S., Pacheco, H., Tarek, N., O'Carroll, D. M., & Allam, N. K. (2024). Untapped potential of scrap brass alloy: a new frontier in the use of brass based photocathodes for stable and durable photoelectrochemical water splitting. In Energy Advances (Vol. 3, Issue 2, pp. 430–441). Royal Society of Chemistry (RSC). https://doi.org/10.1039/d3ya00534h
5. **Abodouh, M. M.,** Khedr, G. E. & Allam, N. K. (2024). Optimizing diamond's electronic band structure via defect engineering for enhanced HER and OER catalysis. International Journal of Hydrogen Energy vol. 61 922–933 (2024). https://doi.org/10.1016/j.ijhydene.2024.03.008
6. Hkiri, K., Mohamed, H. E. A., **Abodouh, M. M.** & Maaza, M. Experimental and theoretical insights into the adsorption mechanism of methylene blue on the (002) WO3 surface. Scientific Reports vol. 14 (2024). Nature https://doi.org/10.1038/s41598-024-78491-3
7. Mara Jezernik Nigel Patterson, Julian Rosas, Mariam Awara, Benjamin Maxwell, Essam Elsahwi, Cynthia A. Rice, **Mohamed Mahrous Abodouh**, In-Operando EIS Monitoring of a PEM Fuel Cell Stack to Understand the Influence of Different Operating Conditions on Stack Performance. ECE conference, https://ecs.confex.com/ecs/248/meetingapp.cgi/Paper/211008